SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF THE MEDICINAL PRODUCT
MONOCLOX 250mg hard gelatin capsules

2. QUALITATIVE AND QUANTITATIVE COMPOSITION
Each capsule contains 250mg of cloxacillin as cloxacillin sodium monohydrate.

Excipients with known effect: sunset yellow, sodium.
Each capsule contains 0.16 mg sunset yellow and 13.2 mg of sodium.
For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM
Hard gelatine Capsule
Orange/orange printed “Medochemie” hard gelatin capsules for oral administration

4. CLINICAL PARTICULARS
4.1. Therapeutic indications
Monoclox is indicated for the treatment of infections caused by gram-positive microorganisms including infections caused by penicillin-resistant staphylococci:-
- Skin and soft tissue infections: boils, abscesses, carbuncles, furunculosis, infected wounds, infected burns
- Infected skin conditions like ulcers, eczema or acne
- Respiratory tract infections
- Osteomyelitis, enteritis, endocarditis
- UTI

4.2. Posology and method of administration
Posology
The usual dose of cloxacillin is the equivalent of 500mg four times daily by mouth or 250mg in 1.5ml of water for injection or if necessary 0.5% lignocaine hydrochloride solution, by intramuscular injection every 4 to 6 hours. The equivalent of 500mg in 10ml of water for injections may be given by slow intravenous injection over 3 to 4 minutes every 4 to 6 hours or by intravenous infusion.

Cloxacillin has been administered by intra-articular injection, dissolved if necessary in a 0.5% solution of lignocaine hydrochloride, and by intraplueral injection in doses of 500mg daily. Using powder for injection, up to 250mg has been dissolved in 3ml of sterile water and inhaled by nebuliser four times daily.
Doses may be doubled in severe infections.

Children up to 2 years of age may be given one-quarter the adult dose and those aged 2 to 10 years, one-half the adult dose.

In staphylococcal meningitis systemic treatment has been supplemented with intrathecal injections of cloxacillin. The usual daily intrathecal does is the equivalent of 5mg for infants from birth up to 2 years of age. 10mg for children up to 12 years of age and 20mg for adults.
The dose should be dissolved in sodium chloride injection 0.9% immediately before use to give a solution containing 10mg per ml.

Method of administration
Oral administration.
This medicine should preferably be taken half an hour before meals. The capsules should be swallowed whole, with a glass of water.

4.3. Contraindications

- Hypersensitivity to the active substance, antibiotics belonging to the beta-lactam family (penicillins, cephalosporins) or to any of the excipients listed in section 6.1;
- **Children under 6 years, as this formulation is not safe for use in this age group.**

4.4. Special warnings and precautions for use

- The occurrence of any allergic manifestation requires discontinuation of the treatment and appropriate therapy initiation.
- Severe hypersensitivity (anaphylaxis) and sometimes fatal reactions have exceptionally been observed in patients treated with beta-lactam antibiotics. Before initiating therapy with cloxacillin, careful enquiry should be made concerning previous hypersensitivity reactions to penicillins and cephalosporins. In 5-10% of cases, allergy to penicillins is crossed with allergy to cephalosporins. In patients with a known history of allergy to penicillins and cephalosporins, cloxacillin is contraindicated.
- Pseudomembranous colitis has been reported with nearly all antibacterial agents including cloxacillin. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea during or subsequent to the administration of any antibiotics. In this situation, adequate therapeutic measures should be initiated immediately. Discontinuation of antibiotic therapy should be considered. Medicinal products inhibiting peristalsis are contra-indicated in this situation (see section 4.8).
- In renal impairment, dose adjustment is required if the creatinine clearance is below 30 ml / min (see section 4.2).
- In cases of liver impairment associated with renal impairment, regardless of the degree of renal insufficiency, dose adjustment is required (see section 4.2).
- Neurological reactions may occur when high doses of penicillin M are given to patients with severe kidney insufficiency or in patients with predisposing factors such as history of seizures, with treated epilepsy or with meningitis (see section 4.8).
- This medicine should not be used in association with methrotrexate (see section 4.5).

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Monoclox, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Monoclox must be discontinued immediately and appropriate alternative therapy instituted.

This medicine contains 13.2mg of sodium per capsule. To be taken into consideration by patients on a controlled sodium diet.

Monoclox also contains sunset yellow which may cause allergic reactions.

4.5. Interaction with other medicinal products and other forms of interaction
Association not recommended: Methrotrexate
Cloxacillin may enhance significantly the blood toxicity of methotrexate due to inhibition of the renal secretion of methotrexate by penicillins.

INR imbalance
Many cases of increased activity of oral anticoagulants were reported in patients receiving antibiotics. The marked infectious or inflammatory context, the patient's age and general condition of the patient are risk factors. Under these circumstances, it seems difficult to distinguish between the infectious pathology and its treatment in the occurrence of the INR imbalance. However, certain classes of antibiotics are more involved: in particular, fluoroquinolones, macrolides, tetracyclines, co-trimoxazole

and certain cephalosporins.

4.6. Fertility, pregnancy and lactation

Pregnancy

Cloxacillin may be used during pregnancy if necessary, as the clinical data on a limited number of patients and the pre-clinical data did not reveal any malformative or foetotoxic effect of penicillins.

Breast-feeding

The passage of penicillins in human milk is low and feed intake far below neonatal therapeutical doses. Therefore breast-feeding is possible during the treatment with this antibiotic.
However, breastfeeding discontinuation (or medicine discontinuation) is advisable on the occurrence of diarrhea, candida or rash in the infant.

4.7. Effects on ability to drive and use machines

Monoclox has no influence on the ability to drive and use machines.

4.8. Undesirable effects

Immune system disorders

Urticaria, Quincke's edema, exceptionally anaphylactic shock (see section 4.4)

Skin and subcutaneous tissue disorders

Maculopapular rashes of allergic or non-allergic origin. DRESS syndrome (drug hypersensitivity syndrome with eosinophilia and systemic symptoms).
Isolated cases of erythroderma and severe bullous drug eruption (erythema multiforme, Stevens Johnson syndrome, Lyell syndrome).

Gastrointestinal disorders

Nausea, vomiting, diarrhea.
Rare cases of pseudomembranous colitis have been reported (see section 4.4)

Hepatobiliary disorders

Rare and moderate increase in transaminases (ASAT and ALAT), exceptionally cholestatic hepatitis.

Nervous system disorders

Beta-lactams are at risk of encephalopathy (confusion, disturbances of consciousness, epilepsy or abnormal movements) and, particularly, in the event of overdose or renal failure.

Renal and urinary disorders

Acute immune-allergic interstitial nephropathies, acute renal failure.

Blood and lymphatic system disorders

Reversible hematological disorders: anemia, thrombocytopenia, leukopenia, neutropenia and agranulocytosis.

General disorders and administration site conditions

Fever

4.9. Overdose

Symptoms

Renal, neuropsychological and digestive symptoms have been reported following overdosage with penicillines M.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Beta-lactamase resistant penicillin's, Antibacterials for systemic use
ATC code: J01CF02

Cloxacillin is an antibiotic in the class of beta-lactam antibiotics of the penicillin group M.

Spectrum of antibacterial activity

Breakpoints

The minimum inhibitory concentrations (MIC) established by the European Committee on Antimicrobial Susceptibility Testing (EUCAST) are presented below:

Breakpoints established by the EUCAST for cloxacillin (2010-04-27, v.1.1)		
Organismes	**Sensitive (S) (mg/l)**	**Resistant (R) (mg/l)**
Staphylococcus aureus	≤ 2	> 2
Staphylococcus lugdunensis	≤ 2	> 2
Staphylocoques coagulase négative	≤ 0.25	**>** 0.25

The prevalence of acquired resistance may vary according to geography and the weather for certain species. It is therefore useful to have information on local data resistance prevalence, especially for the treatment of severe infections. If necessary, it is desirable to obtain a specilized advice mainly when the interest of the drug in some infections can be set due to the level of local prevalence of resistance.

Categories
<u>**COMMONLY SUSCEPTIBLE SPECIES**</u> **Aerobic Gram-positive bacteria** Streptococcus pyogenes **Anaerobes** Clostridium perfringers
INCONSTANT SUSCEPTIBLE SPECIES (**ACQUIRED RESISTANCE > 10%**) **Aerobic Gram-positive bacteria** Staphylococcus aureus (1) Coagulase-negative staphylococci (+)

(+) The prevalence of bacterial resistance is ≥ 50% in France.
(1) The methicillin resistance rate is about 20 to 30% in Staphylococcus aureus and it is found mainly in hospitals.

5.2 Pharmacokinetic properties

<u>Absorption</u>

Cloxacillin is stable in gastric environment. It is well absorbed by the digestive mucous (70 %).

<u>Distribution</u>

- After oral administration, peak plasma concentration is 9mg/l for a dose of 500mg, it is achieved after 1 hour and it is proportional to the dose administered.
- After intravenous injection of 2 g perfusion over 20 minutes, the peak plasma concentration at the end of the perfusion is 280 mg/l.
- Half-life is about 45 minutes in patients with normal renal functions.
- Plasma protein bounding is approximately 90%.
- The product diffuses into the amniotic fluid, fetal blood, synovial fluid and bone tissue.

<u>Biotransformation</u>

Cloxacillin is not metabolised.

Elimination

After oral administration, the non-absorbed fraction is eliminated by the intestinal route in an inactive form. The absorbed fraction is eliminated in an active form essentially through urine and about 10% through the bile duct.

After administration by injection, the elimination is:

- Urinary, in an active form, in 6 hours, approximately 70 to 80% of the injected dose.
- Biliary, in an active form, 20 to 30% of the injected dose.

5.3 Preclinical safety data

Not applicable.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Capsule filling: magnesium stearate

Capsule shell: gelatin, titanium dioxide (E171) and sunset yellow (E110).

6.2 Incompatibilities

None

6.3 Shelf life

24 months

6.4 Special precautions for storage

Store below 30°C, in the original package.

6.5 Nature and contents of container

The capsules are packed in Tropicalized blisters. Available in pack 1000 capsules. Not all pack sizes may be marketed.

6.6 Special precautions for disposal

No special requirements for disposal.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Not all pack sizes may be marketed.

7. MARKETING AUTHORISATION HOLDER

Komedic Sdn. Bhd., No. 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor, Malaysia

8. MANUFACTURER

Medochemie Ltd, (Factory B), 48 Iapetuo Street, Agios Athanassios Industrial Area, 4101 Agios Athanassios, Limassol, CYprus

9. MARKETING AUTHORISATION NUMBER

MAL19871957AZ

10. DATE OF FIRST AUTHORISATION/RENEWAL OF AUTHORISATION

07/1987

11. DATE OF REVISION OF THE TEXT

03/2021